

Measuring Validator Utility in Generate–Verify–Repair NetOps Pipelines: a Confirmatory Paired Study with Shared Initial Candidates

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory study (AC-2026-03) to evaluate whether a multidimensional validator metric suite predicts end-to-end agentic NetOps success better than a single verifier pass rate. Using a synthetic intent→configuration generator and a hosted generate–validate–repair (GVR) adapter under shared_initial_candidate semantics, we ran 200 paired tasks (one shared initial generation per task reused for baseline and guarded). The deterministic validator (deterministic_netops_validator_v5, v5.0) judged every sampled initial candidate valid; no repairs were applied. Baseline = 200/200, guarded = 200/200, paired difference = 0.00 (bootstrap 95% CI [0.00, 0.00], exact McNemar two-sided $p = 1.0$; bootstrap_seed = 1729, resamples = 10000). The observed ceiling invalidated the preregistered power assumptions (targeted 10 percentage-point paired improvement) and rendered the primary confirmatory test uninformative for the originally planned effect. We report preregistered design choices, precise execution accounting, representative validator-trace excerpts, quantitative analysis, failure-mode diagnostics, and artifact-grounded prescriptions for follow-up preregistered experiments (fault injection, multi-sample generation, multi-validator comparison) required to estimate validator coverage and repair value.

I. INTRODUCTION

Automating NetOps workflows with generative models requires reliable mechanisms to avoid harmful, silent, or wrong-but-plausible actions. A common operational pattern is generate–verify–repair (GVR): a generator (LLM) proposes a candidate configuration or plan, a deterministic validator mechanically checks correctness and policy constraints, and, when necessary, a repair module synthesizes corrected candidates or triggers human review. Prior demonstrations of GVR-like architectures in code and systems motivate their adoption in NetOps, but systematic, preregistered measurements of validator coverage (false-positive/false-negative rates), detection latency, and predictive usefulness for end-to-end guarded success are scarce [1].

We preregistered study AC-2026-03 with a paired design that isolates the effect of downstream validator-triggered repair under shared_initial_candidate semantics: for each task the generator produced a single initial candidate that was reused by both the baseline (no repair) and guarded (validator+repair permitted) conditions. Under these semantics any paired difference can only arise down-

stream from validator-triggered repair, not from independent generator sampling. The primary estimand was the paired_success_rate_difference_guarded_minus_baseline and the preregistered confirmatory hypothesis (H1) expected that a multidimensional validator-metric suite would explain more variance in end-to-end success than a single verifier pass rate. Our main contribution is a fully preregistered, artifact-archived experiment and an exact, transparent report of a degenerate confirmatory outcome (ceiling) with concrete, artifact-grounded prescriptions for follow-up designs that can measure validator utility.

II. RELATED WORK

Work across code, systems, and NetOps communities has proposed and prototyped generate–verify–repair or closed-loop verifier architectures to reduce stochastic generator error and enable deterministic acceptance criteria [1]. Recent NetOps and AIOps benchmark efforts document substantive operational failure modes (ticket noise, false premises, and wrong-but-plausible outputs) that complicate end-to-end automation and motivate insertion of verifier layers [2],[3]. Independent system reports and preprints propose self-healing orchestrators, auditable decision traces, and verifier-guided repair to reduce silent or action-triggering failures in tool-augmented LLM systems [4]. Surveys of LLMs for telecommunications and NetOps discuss adaptation and verification trade-offs and motivate metricized evaluation of verification cost versus nominal correctness [5].

These verified records motivate a measurement focus on validator soundness and operational impact but do not provide a standardized, empirically estimable validator-metric suite tailored to NetOps GVR flows. The present preregistered experiment was designed to begin filling that measurement gap under strict preregistration and archived-execution practices; when interpreted against the cited literature we treat prior demonstrations as architectural and motivational rather than as establishing empirical coverage for NetOps validators.

III. METHODOLOGY

Preregistered design and confirmatory intent. We preregistered study AC-2026-03 as a paired confirmatory

experiment to isolate downstream validator/repair effects from generator variability. The registry specified `adapter_family = hosted_netops_gvr_v1` and `generation_semantics = shared_initial_candidate` with `initial_generation_calls_per_task = 1`. The primary estimand was the `paired_success_rate_difference_guarded_minus_baseline`; the primary test was an exact two-sided McNemar with paired bootstrap percentile confidence intervals (`bootstrap_resamples = 10000`, `bootstrap_seed = 1729`). The design targeted detection of an absolute 0.10 paired improvement ($\alpha = 0.05$, ~80% power) under conservative planning assumptions and a preregistered sample size of `task_count = 200` pairs (`planned_episode_count = 400`).

Task generation and constraints. Tasks were drawn deterministically from `deterministic_netops_task_generator_v5` (`task_family = intent_configuration_repair_v1`). Each task manifest entry contains: an `initial_state`, `expected_final_state`, a human-readable intent string, `allowed_fields` and `allowed_touched_fields`, `explicit_workflow_policies` (e.g., `minimum_active_in_groups`, `must_be_down_for_fields`), and a `required_command_sequence`. Task selection followed the preregistered deterministic index rule `challenging_workflow_stress_v1` to produce a stress-profiled suite (levels labeled 'very_hard' and 'extreme' in difficulty fields). Contamination and exclusion rules were preregistered: identical SHA-256 duplicates are excluded and replaced deterministically; near-duplicate tasks (embedding cosine > 0.95) are deterministically excluded and replaced. The execution/analysis artifacts (`analysis/contamination_summary.csv`) record no contamination exclusions.

Model, sampling, and provider accounting. The declared model in `execution/model_configuration.json` is "gpt-5-mini" and the provider bundle is recorded as "openai_responses". Preregistered `model_scope = gpt-5-mini` and `maximum_model_calls = 400`. The adapter executed exactly one shared initial generation per task (`model_calls_used = 200`, as recorded in the execution manifest). Important artifact-grounded sampling detail: `execution/model_configuration.json` records `temperature = null` (`temperature = null/unset`). Explicitly, `null/unset` is not evidence of deterministic hosted-model sampling and does not imply `temperature = 0`; provider-side sampling behavior may remain stochastic and is captured in `provider-call` audit fields. Deterministic reconciliation documents `hosted_model_call_event_count = 200` and `stage_counts.shared_initial_generation = 200`, confirming the single shared generation per pair semantics.

Validator, scoring rule, and repair policy. The pipeline used `deterministic_netops_validator_v5` (`validator_version = 5.0`) as the mechanistic validator. The validator implements rule-based intent-constraint checks and emits per-episode fields used by scoring: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_count` and `violation_codes`, `valid` (boolean), `repair_applied` (boolean), and `repair_provider_trace` fields when a repair is invoked. The preregistered binary scoring

rule `full_intent_constraint_validation` maps `validator.valid` (true/false) to the confirmatory success label. The pipeline permitted at most one repair call when the validator rejected a candidate (`repair_budget_per_episode = 1`); repair events would be logged with `repair_applied = true` and `repair_provider_trace` populated. In practice, the archive shows `repair_applied = false` for all episodes and null repair trace fields.

Execution-level controls, exclusions, and missingness. Operational execution constraints were preregistered: `maximum_attempts_per_call = 1` (no manual retries), deterministic replacement indices for excluded tasks, and a complete-pair primary analysis rule (`paired_binary_analysis_v1` uses only complete pairs). The manifested execution ran to completion with `completed_episode_count = 400`, `failed_episode_count = 0`, and `complete_pair_count = 200`. Missingness artifacts (`analysis/missingness_summary.csv`) show no baseline-only, guarded-only, or both-missing pairs. All provider-call, validator-trace, and raw-response artifacts were archived for reproducibility.

Analysis plan and inferential checks. The primary confirmatory test was the exact McNemar two-sided procedure applied to the paired contingency (`n_11`, `n_10`, `n_01`, `n_00`) with bootstrap percentile 95% confidence intervals (10,000 resamples, `seed = 1729`) for the paired success-rate difference. Secondary preregistered analyses included: per-validator FP-rate and FN-rate estimation (averaged across tasks), mean detection latency per validator, and a linear model predicting a simulated operational-impact score from the validator metric vectors (report adjusted R^2). Multiplicity and exploratory analyses were declared: the primary estimand was unique and exploratory subgroup/error-class analyses would control false discovery rate when multiple p-values were reported. The preregistration also specified sensitivity checks: treating missing or incomplete episodes as failures (zero) and bootstrap stability checks (1,000 resamples recommended during planning; final confirmatory bootstrap used 10,000 resamples as recorded).

Why these choices. The paired `shared_initial_candidate` design was chosen to provide a clean scientific isolation: any improvement observed in the guarded condition must arise from deterministic validator detection and repair (or from repair-induced acceptance differences), not from independent generator sampling differences. The preregistered contamination rules, single-attempt policy, and deterministic task indexing were selected to reduce confounds and keep the confirmatory estimand interpretable. The trade-off is explicit and documented: `shared_initial_candidate` deliberately suppresses generator variability and therefore increases the risk of ceiling/floor outcomes when the single sampled candidate systematically agrees with the validator, a risk that materialized in the present run.

Archive and artifact linkage for reproducibility of methods. All method-level configuration objects and runtime traces referenced above are present in the archived execution bundle: `execution/model_configuration.json` (`declared_model_version = "gpt-5-mini"`, `temperature = null`, `maximum_attempts_per_call = 1`), `execution/task_manifest.jsonl` (determinis-

tic_netops_task_generator_v5 payloads and required sequences), execution/raw_results.jsonl (raw model responses), execution/scoring/*.json (per-episode validator traces), and analysis artifacts (analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/missingness_summary.csv). The preregistration SHA-256 is recorded in the execution manifest (preregistration_sha256 = de37b485421cb0895a98df38b6b3baad1dc7c13c81065e3d8240d2a94a0ff844). These artifacts support independent verification of method-level choices, confirm that the single-sample-per-task and shared_initial_candidate semantics were enforced, and document the validator interface used to produce the binary success labels.

IV. RESULTS

Execution accounting and primary outcomes. The adapter completed the run exactly as preregistered: planned_episode_count = 400, completed_episode_count = 400, failed_episode_count = 0, and model_calls_used = 200 (one shared initial generation per pair). Provider-call audit: hosted_model_call_event_count = 200, completed_hosted_model_call_event_count = 200, cache_hit_count = 0, cache_miss_count = 200, stage_counts.shared_initial_generation = 200. No contamination exclusions or near-duplicate exclusions occurred.

Primary confirmatory result. Every sampled shared initial candidate passed the deterministic validator and no repairs were applied in any episode. Analysis artifacts report baseline_success_count = 200/200 (success_rate = 1.0) and guarded_success_count = 200/200 (success_rate = 1.0). The paired contingency (analysis/paired_contingency_table.csv) is n₁₁ = 200, n₁₀ = 0, n₀₁ = 0, n₀₀ = 0. The paired_success_rate_difference_guarded_minus_baseline = 0.00. Exact McNemar two-sided test p-value = 1.0. Bootstrap percentile 95% confidence interval (10,000 resamples, seed = 1729) = [0.00, 0.00]. The numeric artifacts (analysis/condition_summary.csv, analysis/paired_contingency_table.csv, analysis/analysis_log.jsonl) match these reported values.

Representative validator-trace excerpt (artifact-grounded). To illustrate validator outputs and why no repairs were triggered, we include verbatim key fields from execution/scoring/task-000004-guarded.json (pair-000004):

```
pair_id: pair-000004 normalized_configuration: "interface
edge2 admin down\ninterface edge2 vlan 40\ninterface
edge2 admin up\ninterface edge1 admin down"
parsed_command_count: 4 required_command_count: 4
observed_touched_field_count: 3 valid: true repair_applied:
false validator_id: deterministic_netops_validator_v5
validator_version: 5.0
```

Equivalent validator_trace.validation_after.valid = true and repair_applied = false hold for every archived episode; repair_provider_trace fields are null for all episodes. Because baseline and guarded reused the identical shared_initial_candidate per pair, these per-episode validator

outputs explain the degenerate paired outcome: there were no validator-detected mechanical defects to correct.

Secondary and exploratory diagnostics. Per-validator FP/FN estimates are degenerate given no observed rejections; detection-latency and repair-invocation distributions are uninformative because repair_applied = false for all episodes. The execution bundle contains full per-episode traces and raw responses (execution/raw_results.jsonl) to support reanalysis under alternative scoring or injected-fault conditions. Contamination and missingness artifacts confirm no missing or excluded episodes (analysis/missingness_summary.csv: complete_pairs = 200).

Artifact-grounded diagnostic interpretation. Under shared_initial_candidate semantics any paired difference could only arise from validator-triggered repair. The degenerate all-valid outcome is explained by three artifact-grounded determinants visible in the archive: (1) the deterministic task generator produced candidates that satisfied the validator’s mechanistic constraints for the recorded generation seeds; (2) the single-sample-per-task sampling regime removed generator variability as a source of disagreements; (3) mechanistic validators detect structural and policy violations but can be blind to semantic intent misalignment (correct-by-structure yet incorrect-by-intent). These jointly explain why no repairs occurred and the confirmatory test could not detect the preregistered 10-percentage-point improvement target.

V. DISCUSSION

Design implications of shared_initial_candidate semantics. The preregistered shared_initial_candidate choice intentionally isolates downstream validator and repair effects: because baseline and guarded evaluate the same generator output for each task, any paired improvement must originate from validator-triggered modification or repair rather than from independent generator variance. This property makes the paired estimand a precise probe of repair utility but also concentrates a single-sample ceiling risk: if the shared draw aligns with the validator for most tasks, the design yields no observable discordant pairs. The archive documents exactly this outcome (model_calls_used = 200, one shared generation per pair, and n₁₁ = 200), so the null paired difference is a direct, design-consistent artifact rather than an execution anomaly.

Validator scope and a compact taxonomy. Artifact-grounded traces (validator_id = deterministic_netops_validator_v5, validator_version = 5.0) show the validator enforces mechanistic, checkable properties: syntactic/format correctness, presence of required commands (required_command_count), permitted-field constraints (allowed_touched_fields), and explicit workflow policies (e.g., minimum_active_in_groups). We use a pragmatic taxonomy to reason about coverage and blind spots: (A) syntactic/format checks, (B) structural/ordering constraints (required sequences), (C) policy/domain constraints (protected interfaces, group minima), and (D) semantic intent alignment (whether the produced sequence actually implements the operator intent beyond mechanical constraints). The archived outputs show high coverage for A–C under the synthetic bank

and generation seeds used; category D remains the primary open detection challenge because it requires either richer intent models, supplemental semantic validators, or human adjudication.

Measurement strategies that follow directly from the archive. The artifact bundle supports concrete, experimentally controlled measurement protocols to make validator FP/FN estimable while preserving preregistration rigor. Two complementary, artifact-grounded strategies are: 1) Deterministic fault-injection (recommended preregistered arm): inject known, labeled faults into a controlled fraction of tasks (syntactic mistakes, omitted required commands, swapped intent labels, or policy violations) so that baseline invalidity is non-zero. Because injections are deterministic and recorded in task payloads, the experimenter can compute validator true-positive and false-negative rates against the injected ground truth without human adjudication. This is consistent with our earlier prescription and is supported by the stored `task_manifest` entries and generator seeds. 2) Multi-sample reintroduction of generator variance: adopt `initial_generation_calls_per_task = k >= 3` independent draws per task with recorded seeds. Allow the guarded pipeline to accept any validator-approved draw or to perform repair on a rejected draw. This reintroduces generator-side variability and produces observable discordant pairs when some draws fail validator checks while others pass. The present archive shows that removing generator variance ($k = 1$ shared draw) can mask potential repair utility; switching to multi-draw sampling is a direct remedy. Both strategies are implementable within the same hosted `netops_gvr_v1` adapter semantics and are therefore immediately reproducible from the archived infrastructure.

Statistical and inferential considerations. The paired design isolates repair effects but relies on baseline invalidity or generator variance to provide statistical power for detecting repair utility. Our preregistered power target (detecting a 0.10 absolute paired improvement at ~80% power) assumed a non-negligible baseline failure rate; the observed ceiling (`baseline_success_rate = 1.0`) violates that assumption and renders the primary test uninformative. Artifact-grounded remedies include increasing sample size only when baseline failure probability is non-zero, or altering the design to ensure a controlled non-zero baseline invalidity (for example by deterministic fault injection). Additionally, using multi-draw sampling reintroduces within-pair discordance and increases the potential for detecting repair benefits without requiring impractically large n .

Design trade-offs in validator strictness and operational cost. Mechanistic validators that are stricter on structural or policy checks reduce the risk of silent mechanical errors but may increase false positives (rejecting acceptable variants) and thus trigger more repairs or human review. Conversely, permissive validators reduce repair invocation but risk undetected defects. The appropriate operating point depends on operational cost trade-offs (verification cost, human review burden, repair latency). While our execution did not observe repairs and thus cannot measure these trade-offs empirically, the execu-

tion artifacts (including per-episode `parsed_command_count`, `required_command_count`, and `repair_applied` flags) form the necessary raw data fields for future experiments that instrument and quantify repair cost and human-review burden under alternative validator thresholds.

Practical orchestration and human-in-the-loop designs. The archive suggests pragmatic deployment patterns to mitigate semantic-misalignment risk (category D above) even when mechanistic validators perform well: (1) multi-validator arbitration (cross-checks using two independent validators with different coverage profiles), (2) gated human-in-the-loop acceptance for changes flagged as semantically risky, and (3) auditable decision traces that preserve the initial candidate, validator results, and any repair prompts/outputs for operator review. These are not measured outcomes in this run but are operationally plausible mitigations that can be evaluated empirically using the archived adapter and task bank.

How the archived artifacts support reproducible next steps. All claims above are anchored to explicit artifacts in the bundle: `task_manifest` entries (deterministic task generator seeds and required sequences), `execution/model_configuration.json` (declared model = `gpt-5-mini` and temperature = null), per-episode validator traces (`validation_before/after`, `valid` flags), and analysis artifacts (`paired_contingency_table.csv`, `condition_summary.csv`). Because these artifacts encode both the deterministic seeds and the full validator observations, they permit exact replication of the present ceiling outcome and straightforward extensions (inject faults, increase k , add validators) that preserve preregistration and reproducibility.

Threats to validity (brief restatement). Internal validity: the `shared_initial_candidate` semantics intentionally isolate repair effects but increase single-sample ceiling risk. Construct validity: `validator.valid` operationalizes mechanistic correctness but may not capture semantic intent misalignment. External validity: synthetic tasks, a single model (`gpt-5-mini`), and a single deterministic validator limit generalization to heterogeneous production networks and alternative model/validator designs. Conclusion validity: the degenerate outcome produces no variance for the primary estimand, so inferential conclusions about validator benefit are not possible from this execution alone. These limitations are explicit in the archived manifests and motivate the artifact-grounded follow-up arms described above.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics constrained inference: baseline and guarded reused the same initial candidate per pair; any paired difference could only arise from validator-triggered repair (not from independent generator sampling).
- Synthetic task bank (difficulty `'very_hard'/'extreme'`) is not equivalent to heterogeneous production networks (multi-vendor devices, real ticket noise); external validity is limited.
- Single-model experiment: only `gpt-5-mini` was executed; no multi-model generality claims are made.

- No repairs were applied because all initial candidates passed the deterministic validator; therefore the study could not estimate repair effectiveness or validator false-negative behavior in this execution.
- Validator scope: `deterministic_netops_validator_v5` enforces mechanistic intent-constraint checks but does not guarantee detection of semantic intent misalignment (correct-by-structure but incorrect-by-intent).
- Recorded `execution/model_configuration.json` `temperature = null/unset` does not imply deterministic sampling; provider-side sampling behavior may vary and is documented in execution manifests.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory study (AC-2026-03) under `shared_initial_candidate` semantics to measure validator utility in NetOps GVR pipelines. For the synthetic task bank, the sampled `gpt-5-mini` generations, and `deterministic_netops_validator_v5` used here, every sampled initial candidate passed the validator’s mechanistic checks so no repairs were applied and guarded success equaled baseline (200/200). The preregistered power assumptions (targeted 10-percentage-point improvement) were invalidated by this ceiling outcome. The result is artifact-grounded and reproducible from the archived bundle. We publish the exact execution and analysis artifacts to support replication and provide concrete, preregistered experiment prescriptions (fault injection, multi-sample generation, multi-validator comparison) that are required to produce estimable validator FP/FN behavior and repair value in operationally meaningful conditions.

DISCLOSURE STATEMENT

Disclosure (concise): Declared model and provider: `gpt-5-mini` via provider bundle `'openai_responses'`. Orchestration/adaptor: `hosted_netops_gvr_v1`. Deterministic validator: `deterministic_netops_validator_v5` (`validator_version = 5.0`). Preregistration: AC-2026-03 (`preregistration_sha256 = de37b485421cb0895a98df38b6b3baad1dc7c13c81065e3d8240d2a94a0ff844`). Master prompt immutable reference: `provenance/master_prompt.txt`, SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b39b15ba`. Autonomy boundary: a human Research Director selected the broad topic family and approved the master prompt before the autonomy lock; after the lock the AI system autonomously performed literature review, preregistration, experiment design, execution, analysis, internal peer review and manuscript drafting without manual editing of technical content. Sampling/generation semantics: `shared_initial_candidate` (one initial sampled generation per task reused for baseline and guarded); `execution/model_configuration.json` recorded `temperature = null/unset` (`null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`). Call budget and usage (preregistered): `maximum_model_calls = 400`; `actual_model_calls_used = 200` (one shared initial generation per pair). All raw outputs, per-episode validator traces, execution

manifests, and analysis artifacts are archived in the supplied execution bundle (see `execution/execution_manifest.json`). No human manually edited the manuscript technical content after the autonomy lock.

REFERENCES

- [1] [1] R. Cadenas, "Convergent Correctness in Stochastic Code Generation: A Generate-Verify-Repair Architecture for Deterministic Validation of Autonomous AI Coding Agents in Regulated Environments," *SSRN Electronic Journal*, 2026. doi:10.2139/ssrn.6754899.
- [2] [2] K.-H. Tseng, N. Bogahawatta, Y. Ginige, K. Patel, K. Dakic, S. Seneviratne, "Fault-Bench: Towards Benchmarking Network Troubleshooting LLM Agents under Unreliable User Tickets," *arXiv:2608.27021*, 2026.
- [3] [3] Y. Liu, C. Pei, L. Xu, B. Chen, M. Sun, Z. Zhang, Y. Sun, S. Zhang, K. Wang, H. Zhang, J. Li, G. Xie, X. Wen, X. Nie, M. Ma, P. Dan, "OpsEval: A Comprehensive IT Operations Benchmark Suite for Large Language Models," *arXiv:2310.07637*, 2023.
- [4] [4] R. S. Babu and A. Agrawal, "Self-Healing Agentic Orchestrators for Reliable Tool-Augmented Large Language Model Systems," *arXiv:2606.01416*, 2026.
- [5] [5] H. Zhou, C. Hu, Y. Yuan, Y. Cui, Y. Jin, C. Chen, H. Wu, D. Yuan, L. Jiang, D. Wu, X. Liu, J. Zhang, X. Wang, J. Liu, "Large Language Model (LLM) for Telecommunications: A Comprehensive Survey on Principles, Key Techniques, and Opportunities," *IEEE Commun. Surveys & Tutorials*, 2024. doi:10.1109/COMST.2024.3465447.